

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Numerical

Let ABCD be a square of side of unit length.

Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the lines AD and AB are tangent to it, is also drawn. Let a tangent line from the point C to the circle C_2 meet the side AB at E. If the length of EB is $\alpha + \sqrt{3}\beta$, where α, β are integers, then $\alpha + \beta$ is equal to _____

Q2: 16 March (Shift 2) - Single Correct

Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$

($a < 0$) be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then

the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is

equal to :

(1) $\sqrt{11}$

(2) $\sqrt{7}$

(3) $\sqrt{6}$

(4) $\sqrt{10}$

Q3: 17 March (Shift 1) - Single Correct

The line $2x - y + 1 = 0$ is a tangent to the circle at the point (2,5) and the centre of the circle lies on $x - 2y = 4$. Then, the radius of the circle is:

(1) $3\sqrt{5}$

(2) $5\sqrt{3}$

(3) $5\sqrt{4}$

(4) $4\sqrt{5}$

Q4: 17 March (Shift 1) - Single Correct

Choose the incorrect statement about the two circles whose equations are given below:

$x^2 + y^2 - 10x - 10y + 41 = 0$ and

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$$x^2 + y^2 - 16x - 10y + 80 = 0$$

(1) Distance between two centres is the average of radii of both the circles.

(2) Both circles' centres lie inside region of one another.

(3) Both circles pass through the centre of each other.

(4) Circles have two intersection points.

Q5: 17 March (Shift 1) - Numerical

The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$x^2 + y^2 - 24x - 10y + 160 = 0 \text{ is } \underline{\hspace{2cm}}.$$

Q6: 17 March (Shift 2) - Single Correct

Two tangents are drawn from a point P to the circle $x^2 + y^2 - 2x - 4y + 4 = 0$, such that the angle between these tangents is $\tan^{-1}\left(\frac{12}{5}\right)$, where $\tan^{-1}\left(\frac{12}{5}\right) \in (0, \pi)$. If the centre of the circle is denoted by C and these tangents touch the circle at points A and B , then the ratio of the areas of $\triangle PAB$ and $\triangle CAB$ is :

(1) 11: 4

(2) 9: 4

(3) 3: 1

(4) 2: 1

Q7: 18 March (Shift 1) - Single Correct

Choose the correct statement about two circles whose equations are given below:

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$x^2 + y^2 - 22x - 10y + 137 = 0$$

(1) circles have same centre

(2) circles have no meeting point

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(3) circles have only one meeting point

(4) circles have two meeting points

Q8: 18 March (Shift 1) - Single Correct

For the four circles M, N, O and P, following four equations are given :

Circle M : $x^2 + y^2 = 1$

Circle N : $x^2 + y^2 - 2x = 0$

Circle O : $x^2 + y^2 - 2x - 2y + 1 = 0$

Circle P : $x^2 + y^2 - 2y = 0$

If the centre of circle M is joined with centre of the circle N, further centre of circle N is joined with centre of the circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, then these lines form the sides of a :

(1) Rhombus

(2) Square

(3) Rectangle

(4) Parallelogram

Q9: 18 March (Shift 2) - Single Correct

Let $S_1 : x^2 + y^2 = 9$ and $S_2 : (x - 2)^2 + y^2 = 1$ Then the locus of center of a variable circle S which touches S_1 internally and S_2 externally always passes through the points:

(1) $(0, \pm\sqrt{3})$

(2) $\left(\frac{1}{2}, \pm\frac{\sqrt{5}}{2}\right)$

(3) $\left(2, \pm\frac{3}{2}\right)$

(4) $(1, \pm 2)$

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Answer Key

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Q1 (1)

Q2 (3)

Q3 (1)

Q4 (2)

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Q5 (1)

Q6 (2)

Q7 (3)

Q8 (2)

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Q9 (3)

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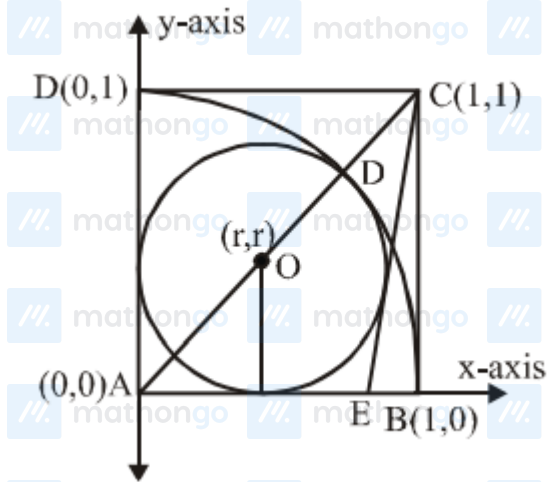
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Q1 (I)



Here $AO + OD = 1$ or $(\sqrt{2} + 1)r = 1$

$$\Rightarrow r = \sqrt{2} - 1$$

equation of circle $(x - r)^2 + (y - r)^2 = r^2$

Equation of CE

$$y - 1 = m(x - 1)$$

$$mx - y + 1 - m = 0$$

It is tangent to circle

$$\therefore \left| \frac{mr - r + 1 - m}{\sqrt{m^2 + 1}} \right| = r$$

$$\left| \frac{(m-1)r + 1 - m}{\sqrt{m^2 + 1}} \right| = r$$

$$\frac{(m-1)^2(r-1)^2}{m^2 + 1} = r^2$$

$$\text{Put } r = \sqrt{2} - 1$$

$$\text{On solving } m = 2 - \sqrt{3}, 2 + \sqrt{3}$$

Taking greater slope of CE as $2 + \sqrt{3}$

$$y - 1 = (2 + \sqrt{3})(x - 1)$$

$$\text{Put } y = 0$$

$$-1 = (2 + \sqrt{3})(x - 1)$$

$$\frac{-1}{2 + \sqrt{3}} \times \left(\frac{2 - \sqrt{3}}{2 - \sqrt{3}} \right) = x - 1$$

$$x - 1 = \sqrt{3} - 1$$

Hints and Solutions

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$$EB = 1 - x = 1 - (\sqrt{3} - 1)$$

$$EB = 2 - \sqrt{3}$$

Q2 (3)

$$x^2 + y^2 + ax + 2ay + c = 0$$

$$2\sqrt{g^2 - c} = 2\sqrt{\frac{a^2}{4} - c} = 2\sqrt{2}$$

$$\Rightarrow \frac{a^2}{4} - c = 2 \dots (1)$$

$$2\sqrt{f^2 - c} = 2\sqrt{a^2 - c} = 2\sqrt{5}$$

$$\Rightarrow a^2 - c = 5$$

(1)&

(2)

$$\frac{3a^2}{4} = 3 \Rightarrow a = -2 \quad (a < 0)$$

$$\therefore c = -1$$

$$\text{Circle} \Rightarrow x^2 + y^2 - 2x - 4y - 1 = 0$$

$$\Rightarrow (x - 1)^2 + (y - 2)^2 = 6$$

$$\text{Given } x + 2y = 0 \Rightarrow m = -\frac{1}{2}$$

$$m_{\text{tangent}} = 2$$

Equation of tangent

$$\Rightarrow (y - 2) = 2(x - 1) \pm \sqrt{6}\sqrt{1 + 4}$$

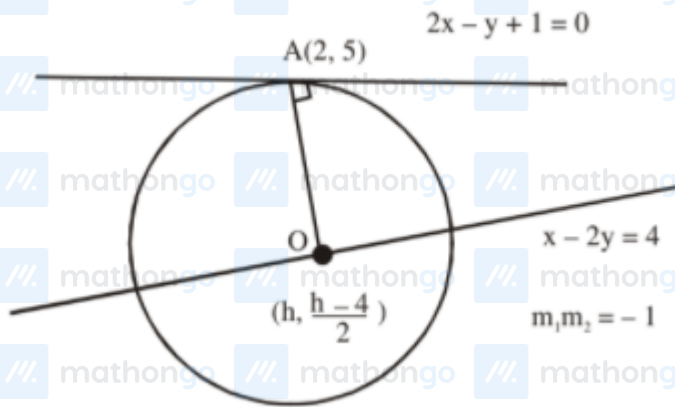
$$\Rightarrow 2x - y \pm \sqrt{30} = 0$$

$$\text{Perpendicular distance from } (0, 0) = \left| \frac{\pm\sqrt{30}}{\sqrt{4+1}} \right| = \sqrt{6}$$

Q3 (1)

Hints and Solutions

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$$\left(\frac{h - \frac{h-4}{2}}{2 - h} \right) (2) = -1$$

$$h = 8$$

center (8, 2)

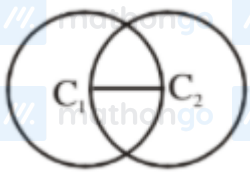
$$\text{Radius} = \sqrt{(8-2)^2 + (2-5)^2} = 3\sqrt{5}$$

Q4 (2)

$$r_1 = 3, c_1(5, 5)$$

$$r_2 = 3, c_2(8, 5)$$

$$C_1 C_2 = 3, r_1 = 3, r_2 = 3$$



Q5 (1)

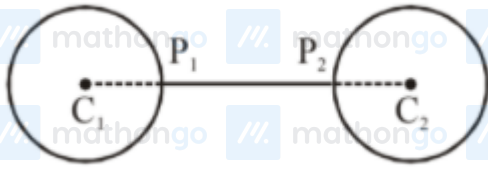
$$\text{Given } C_1(5, 5), r_1 = 3 \text{ and } C_2(12, 5), r_2 = 3$$

$$\text{Now, } C_1 C_2 > r_1 + r_2$$

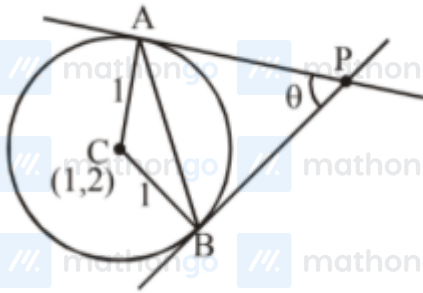
Hints and Solutions

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$$\text{Thus, } (P_1P_2)_{\min} = 7 - 6 = 1$$



Q6 (2)



$$\tan \theta = \frac{12}{5}$$

$$PA = \cot \frac{\theta}{2}$$

$$\therefore \text{area of } \triangle PAB = \frac{1}{2} (PA)^2 \sin \theta = \frac{1}{2} \cot^2 \frac{\theta}{2} \sin \theta$$

$$= \frac{1}{2} \left(\frac{1+\cos \theta}{1-\cos \theta} \right) \sin \theta$$

$$\text{area of } \triangle CAB = \frac{1}{2} \sin \theta = \frac{1}{2} \left(\frac{12}{13} \right) = \frac{6}{13}$$

$$\therefore \frac{\text{area of } \triangle PAB}{\text{area of } \triangle CAB} = \frac{9}{4}$$

Q7 (3)

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$A(5, 5), R_1 = 3$$

$$x^2 + y^2 - 22x - 10y + 137 = 0$$

$$B(11, 5), R_2 = 3$$

$$AB = 6 = R_1 + R_2$$

Touch each other externally

 $\Rightarrow$ circles have only one meeting point.

Hints and Solutions

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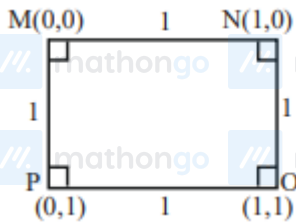
Q8 (2)

$$M : x^2 + y^2 = 1 \quad (0,0)$$

$$N : x^2 + y^2 - 2x = 0 \quad (1,0)$$

$$O : x^2 + y^2 - 2x - 2y + 1 = 0 \quad (1,1)$$

$$P : x^2 + y^2 - 2y = 0 \quad (0,1)$$

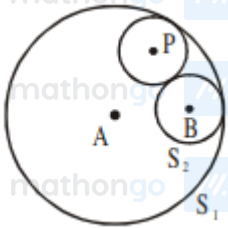


Q9 (3)

$$S_1 : x^2 + y^2 = 9 \quad \begin{matrix} r_1 = 3 \\ A(0,0) \end{matrix}$$

$$S_2 : (x-2)^2 + y^2 = 1 \quad \begin{matrix} r_2 = 1 \\ B(2,0) \end{matrix}$$

$$\therefore C_1 C_2 = r_1 - r_2$$



$\therefore$ given circle are touching internally

Let a variable circle with centre P and radius

r

$$\Rightarrow PA = r_1 - r \text{ and } PB = r_2 + r$$

$$\Rightarrow PA + PB = r_1 + r_2$$

$$\Rightarrow PA + PB = 4 (> AB)$$

Hints and Solutions

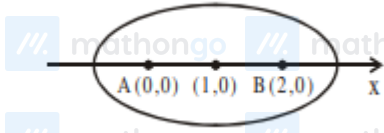
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$\Rightarrow$ Locus of P is an ellipse with foci at A(0, 0) and B(2, 0) and length of major axis is $2a = 4$,

$$e = \frac{1}{2}$$

$\Rightarrow$ centre is at (1,0) and $b^2 = a^2 (1 - e^2) = 3$

if x -ellipse



$$\Rightarrow E : \frac{(x-1)^2}{4} + \frac{y^2}{3} = 1$$

which is satisfied by $\left(2, \pm \frac{3}{2}\right)$